

CYBERSECURITY PORTFOLIO

RESEARCH & ANALYSIS



Common Linux Vulnerabilities

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Portfolio Edition

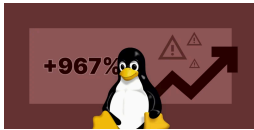
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Common Linux Vulnerabilities



Broad Security Vulnerabilities



Default or no password(s)

- Having a default or missing a password is a critical vulnerability in Linux and other operating systems. A lack of a secure password/no password at all is similar to leaving your front door wide open to a burglar. This could be especially bad if the account with a weak password is an administrative account.

Private Key Exposure

- Having your private key exposed is another critical vulnerability. A private key being stolen allows for bad actors to control and manipulate whatever the key was tied to. An example of this could be a webserver running on Linux.

Service Vulnerabilities

- A classic example of a vulnerability is running an outdated service. An example of this could be utilizing Telnet, or running an old version of a software your organization uses. It is also important to run the latest version of the Linux kernel as well.

Specific Recent and Common Vulnerabilities



Copy Fail (CVE-2026-31431) Recent

- Local privilege escalation
- Exploit activity blends into normal system behavior through standard system calls, making it hard to spot typical monitoring.
- A bug in the “checking” step gets skipped or misused during a copy operation letting a normal user trick the kernel into writing data somewhere it shouldn’t
- An attacker with unprivileged access could use this exploit to escalate to root

Dirty Pipe (CVE-2022-0847) Common

- Dirty pipe is a bug in how the kernel manages the internal buffer for pipes
- A flag that didn't get reset properly a regular user could abuse a pipe to overwrite the contents of files they normally wouldn't have permission to touch
- Some files include those that are owned by root or marked as read only
- Someone could inject their own commands into a root owned script or binary and get it to run as root next time it executes

